

Caleb Woldemedhine

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EDUCATION

University of Denver

B.S. in Computer Science (AI & ML Concentration)

Minors: Mathematics and Statistics

GPA: 3.66

Relevant Coursework: Database Organization & Management, Topics in Artificial Intelligence (Machine Learning & LLMs), Applied Math for Data Science and AI, Machine Learning for IoT and Embedded Devices, Linear Algebra, Introduction to Probability

Denver, CO

Sept 2023 - Jun 2027

EXPERIENCE

Palo Alto Networks

Santa Clara, CA

Software Engineering Intern

June 2026 - Present

- Designed and built an LLM-driven visualization service within a pub/sub multi-agent backend that automatically selects, validates, and shapes agent query results into structured JSON outputs consumed directly by frontend chart and table components, with a deterministic fallback chain ensuring every request returns a renderable response.
- Developed a benchmarking harness achieving 86% chart-selection accuracy across 500+ scenarios, migrated the LLM backend from Anthropic to Vertex AI with retry/backoff logic, and re-architected a fragmented codebase into a catalog-driven design where each component's data contract, validation rules, and selection criteria live in a single spec file.

Frederick S. Pardee Institute for International Futures

Denver, CO

Data Technician

Sept 2025 - Present

- Cleaned and preprocessed international education datasets using Python and pandas, standardizing country-year coverage, resolving missing or inconsistent values, and running quality checks to prepare tables for database imports and website updates.
- Built a Python import script to convert cleaned Excel datasets into SQL-ready inserts, automate formatting and loading workflows, and verify imported values against the Data Dictionary for accurate database integration.

Tempus Automation

Highlands Ranch, CO

Software Engineering Intern

Jun 2025 - Aug 2025

- Trained model using ML.NET and evaluated a C# model to automate deck design generation from computer vision inputs, using a custom dataset of materials, dimensions, pricing rules, and design constraints.
- Built a relational SQL database to store inventory, pricing logic, and model training data, then integrated the trained model into a web platform for real-time design recommendations and dynamic cost estimates.

Denver Center of African Art

Denver, CO

Web Development Intern

Jun 2024 - Sept 2024

- Built a responsive website using HTML, CSS, and JavaScript and integrated Stripe donation payments with receipts and donor record tracking.

Michael Auto Sales

Denver, CO

IT & Data Assistant

Sept 2022 - Jan 2024

- Organized profit, expense, and inventory data using Excel, Python, SQL, and DealerCloud to track business performance and maintain accurate vehicle records.

PROJECTS

Stock Prediction and Analysis Tool

- Built a stock analysis and forecasting website with a Python backend and JavaScript frontend that gives users an interactive dashboard with price forecasts, sentiment insights, trading signals, and model accuracy results powered by machine learning.

DuDraw AI Coder

- Built a full-stack AI website where users enter a prompt and a large language model generates DuDraw code that turns their ideas into interactive drawings and simple games.

NYC COVID-19 Spread Simulation

- Created an interactive COVID-19 spread simulation with adjustable controls such as lockdowns, patient zero, speed and real-time epidemic curve visualization.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, C, C++, C#, SQL, JavaScript, TypeScript, HTML, CSS, React, Next.js, R, Haskell

Libraries & Technologies: PyTorch, TensorFlow, Keras, Scikit-learn, Prophet, Pandas, Plotly, NumPy, Excel, Tableau, REST APIs, AWS, ML.NET, Docker, Git, Linux, JSON

LEADERSHIP

National Society of Black Engineers

Denver, CO

President

Jun 2025 - Present

- Raised \$40,000 to send 35 students to the NSBE National Convention and organized a multi-company hiring fair, leading employer outreach, sponsorships, and event operations to expand student career opportunities.